

Recitation 2

ECON UN3412: Econometrics

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Note: Section titles highlighted in **red** are **supplementary material** that were not covered in class. They may be more mathematically advanced.

1 Ordinary Least Squares (OLS)

Consider the linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

where Y_i is the dependent variable, X_i is the independent variable, and u_i is the unobserved error term.

Given sample data $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$, we want to construct a line of best fit:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

which generates a predicted value $\hat{Y}_i$ for each observation at X_i .

Ordinary Least Squares (OLS) chooses the intercept $\hat{\beta}_0$ and slope $\hat{\beta}_1$ to fit a line through data points such that the **sum of squared vertical residuals (distances)** between the observed points and the fitted line is minimized (see Figure 1):

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i))^2$$

1.1 Geometric Motivation

How do we decide which line is the “best” fit? We can naturally motivate this using the **definition of Euclidean distance** in the Cartesian plane $\mathbb{R}^2$.

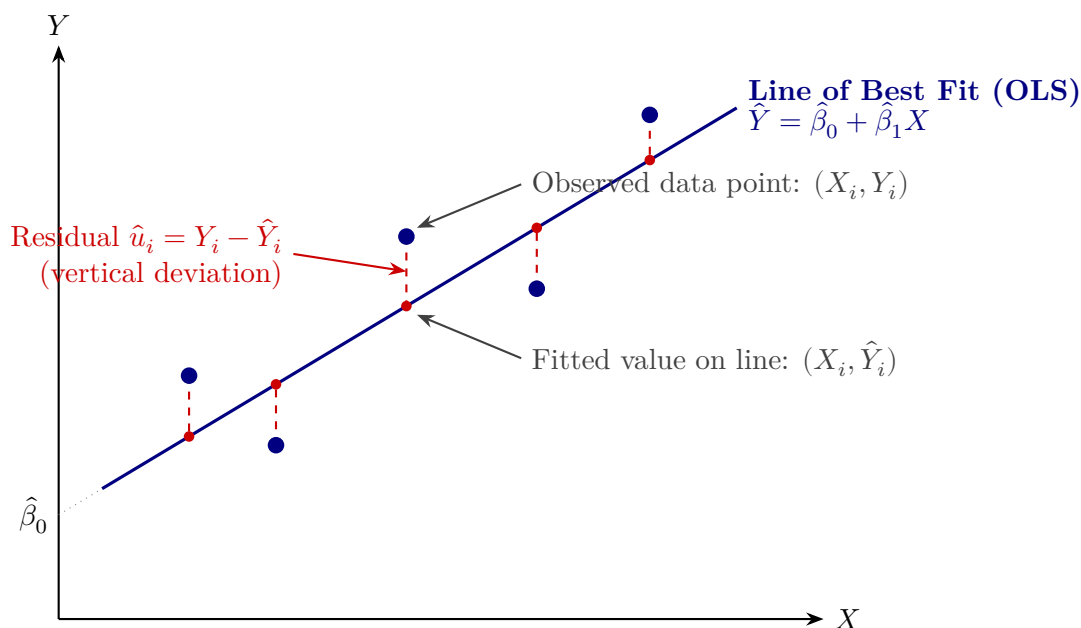


Figure 1: OLS minimizes the sum of squared vertical residuals $\sum_{i=1}^n \hat{u}_i^2$.

1. Distance for a Single Data Point

In 2D coordinate geometry, the Euclidean distance between any two points (x_1, y_1) and (x_2, y_2) is given by the Pythagorean theorem:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

For an observed data point (X_i, Y_i) and its corresponding prediction $(X_i, \hat{Y}_i)$ evaluated at the same X_i , the horizontal deviation is zero ($\Delta X = 0$). Thus, the 2D Euclidean distance between the actual observation and the line is purely **vertical**:

$$d_i = \sqrt{(X_i - X_i)^2 + (Y_i - \hat{Y}_i)^2} = \sqrt{(Y_i - \hat{Y}_i)^2} = |Y_i - \hat{Y}_i| = |\hat{u}_i|$$

where $\hat{u}_i = Y_i - \hat{Y}_i$ is the **residual** (the prediction error for observation i).

The **squared distance** for observation i is therefore:

$$d_i^2 = (Y_i - \hat{Y}_i)^2 = \hat{u}_i^2$$

2. Distance Across the Entire Sample

Across all n data points, the total Euclidean distance between the vector of observed values $\mathbf{Y} = (Y_1, \dots, Y_n)'$ and fitted values $\hat{\mathbf{Y}} = (\hat{Y}_1, \dots, \hat{Y}_n)'$ in n -dimensional space is:

$$\|\mathbf{Y} - \hat{\mathbf{Y}}\|_2 = \sqrt{\sum_{i=1}^n d_i^2} = \sqrt{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}$$

Because the square root function is strictly increasing, **minimizing the total Euclidean distance** between the actual data and the regression line is mathematically identical to **minimizing the Sum of Squared Residuals (SSR)**:

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^n d_i^2 = \min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^n \hat{u}_i^2 = \min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^n (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i))^2$$

2 Derivation of the OLS Estimators

We want to find the values of $(\hat{\beta}_0, \hat{\beta}_1)$ that minimize the Sum of Squared Residuals (SSR), defined as the objective function:

$$S(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$$

2.1 Step 1: First-Order Conditions (OLS Normal Equations)

Setting the partial derivatives of $S(\hat{\beta}_0, \hat{\beta}_1)$ with respect to $\hat{\beta}_0$ and $\hat{\beta}_1$ equal to zero yields the two **OLS Normal Equations**:

1. First Normal Equation (Differentiating with respect to $\hat{\beta}_0$):

Using the chain rule:

$$\frac{\partial S}{\partial \hat{\beta}_0} = \sum_{i=1}^n 2(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)(-1) = -2 \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)$$

Setting $\frac{\partial S}{\partial \hat{\beta}_0} = 0$ and dividing by -2 :

$$\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0 \quad \Leftrightarrow \quad \sum_{i=1}^n \hat{u}_i = 0$$

Key Implication 1: The sum (and therefore the sample mean) of the OLS residuals is always zero: $\bar{\hat{u}} = \frac{1}{n} \sum_{i=1}^n \hat{u}_i = 0$.

2. Second Normal Equation (Differentiating with respect to $\hat{\beta}_1$):

Using the chain rule:

$$\frac{\partial S}{\partial \hat{\beta}_1} = \sum_{i=1}^n 2(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)(-X_i) = -2 \sum_{i=1}^n X_i(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)$$

Setting $\frac{\partial S}{\partial \hat{\beta}_1} = 0$ and dividing by -2 :

$$\sum_{i=1}^n X_i(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0 \quad \Leftrightarrow \quad \sum_{i=1}^n X_i \hat{u}_i = 0$$

Key Implication 2: The regressor X_i and the residuals $\hat{u}_i$ are orthogonal in the sample: their sample covariance is zero ($\sum_{i=1}^n X_i \hat{u}_i = 0$).

2.2 Why Can We Take Derivatives and Set Them to Zero?

Why does setting the gradient equal to zero ($\nabla S(\hat{\beta}_0, \hat{\beta}_1) = 0$) guarantee that we have found the **unique global minimum** of the sum of squared residuals?

1. **Continuous Differentiability (Smoothness):** $S(\hat{\beta}_0, \hat{\beta}_1)$ is a quadratic polynomial in $\hat{\beta}_0$ and $\hat{\beta}_1$. It is continuous and infinitely differentiable everywhere on $\mathbb{R}^2$, meaning there are no sharp kinks, boundaries, or asymptotes.
2. **Unconstrained Optimization:** The parameter choices $\hat{\beta}_0, \hat{\beta}_1 \in \mathbb{R}$ are unconstrained (they can take any real number), so any local extremum must occur at a stationary point where the gradient is zero:

$$\nabla S(\hat{\beta}_0, \hat{\beta}_1) = \begin{pmatrix} \frac{\partial S}{\partial \hat{\beta}_0} \\ \frac{\partial S}{\partial \hat{\beta}_1} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

3. **Strict Convexity (Guaranteed Unique Global Minimum):** To ensure that the critical point is a **minimum** (and not a maximum or saddle point), we inspect the Hessian matrix of second-order partial derivatives:

$$H = \begin{pmatrix} \frac{\partial^2 S}{\partial \hat{\beta}_0^2} & \frac{\partial^2 S}{\partial \hat{\beta}_0 \partial \hat{\beta}_1} \\ \frac{\partial^2 S}{\partial \hat{\beta}_1 \partial \hat{\beta}_0} & \frac{\partial^2 S}{\partial \hat{\beta}_1^2} \end{pmatrix} = 2 \begin{pmatrix} n & \sum_{i=1}^n X_i \\ \sum_{i=1}^n X_i & \sum_{i=1}^n X_i^2 \end{pmatrix}$$

- The first principal minor is $2n > 0$ (for any sample size $n \geq 1$).
- The determinant of H is:

$$\det(H) = 4 \left(n \sum_{i=1}^n X_i^2 - \left(\sum_{i=1}^n X_i \right)^2 \right) = 4n \sum_{i=1}^n (X_i - \bar{X})^2$$

As long as there is sample variation in X ($\sum (X_i - \bar{X})^2 > 0$, i.e., X is not a constant),

$$\det(H) > 0.$$

Since both principal minors are strictly positive, H is **strictly positive definite**. This proves that $S(\hat{\beta}_0, \hat{\beta}_1)$ is **strictly convex**, guaranteeing that the solution to the first-order conditions is the **unique global minimum**.

2.3 Step 2: Solving for the Intercept $\hat{\beta}_0$

From the first FOC:

$$\sum_{i=1}^n Y_i - \sum_{i=1}^n \hat{\beta}_0 - \sum_{i=1}^n \hat{\beta}_1 X_i = 0$$

Since $\sum_{i=1}^n \hat{\beta}_0 = n\hat{\beta}_0$:

$$\sum_{i=1}^n Y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^n X_i = 0$$

Divide the entire equation by the sample size n :

$$\bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{X} = 0$$

Rearranging gives the formula for the OLS intercept estimator:

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}}$$

Geometric Takeaway: The OLS regression line **always passes through the point of sample means** $(\bar{X}, \bar{Y})$.

2.4 Step 3: Solving for the Slope $\hat{\beta}_1$

Now substitute $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$ into the second FOC:

$$\sum_{i=1}^n X_i (Y_i - (\bar{Y} - \hat{\beta}_1 \bar{X}) - \hat{\beta}_1 X_i) = 0$$

Group the terms:

$$\sum_{i=1}^n X_i [(Y_i - \bar{Y}) - \hat{\beta}_1 (X_i - \bar{X})] = 0$$

Distribute X_i :

$$\sum_{i=1}^n X_i (Y_i - \bar{Y}) - \hat{\beta}_1 \sum_{i=1}^n X_i (X_i - \bar{X}) = 0$$

Useful Algebraic Identities

To express the two summation terms in terms of sample deviations $(X_i - \bar{X})$ and $(Y_i - \bar{Y})$, we use a fundamental property of sample means: **the sum of deviations about the mean is always zero**:

$$\sum_{i=1}^n (Y_i - \bar{Y}) = \sum_{i=1}^n Y_i - \sum_{i=1}^n \bar{Y} = n\bar{Y} - n\bar{Y} = 0$$

Using this property, we establish two standard identities:

Identity 1: $\sum_{i=1}^n X_i(Y_i - \bar{Y}) = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$

Proof: Expand the right-hand side by distributing $(X_i - \bar{X})$:

$$\begin{aligned} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) &= \sum_{i=1}^n [X_i(Y_i - \bar{Y}) - \bar{X}(Y_i - \bar{Y})] \\ &= \sum_{i=1}^n X_i(Y_i - \bar{Y}) - \sum_{i=1}^n \bar{X}(Y_i - \bar{Y}) \end{aligned}$$

Because the sample mean $\bar{X}$ is a constant across all observations i , we factor it out of the second summation:

$$\sum_{i=1}^n \bar{X}(Y_i - \bar{Y}) = \bar{X} \underbrace{\sum_{i=1}^n (Y_i - \bar{Y})}_{=0} = \bar{X} \cdot 0 = 0$$

Since the second term vanishes, we obtain:

$$\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n X_i(Y_i - \bar{Y})$$

Identity 2: $\sum_{i=1}^n X_i(X_i - \bar{X}) = \sum_{i=1}^n (X_i - \bar{X})^2$

Proof: This is simply Identity 1 where $Y_i = X_i$. Distributing $(X_i - \bar{X})$ on the right-hand side:

$$\begin{aligned} \sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X}) \\ &= \sum_{i=1}^n X_i(X_i - \bar{X}) - \bar{X} \underbrace{\sum_{i=1}^n (X_i - \bar{X})}_{=0} \\ &= \sum_{i=1}^n X_i(X_i - \bar{X}) - \bar{X} \cdot 0 \\ &= \sum_{i=1}^n X_i(X_i - \bar{X}) \end{aligned}$$

Final Formula for $\hat{\beta}_1$

Substituting these identities into the distributed FOC:

$$\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) - \hat{\beta}_1 \sum_{i=1}^n (X_i - \bar{X})^2 = 0$$

Solving for $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{s_{XY}}{s_X^2} = \frac{\widehat{\text{Cov}}(X, Y)}{\widehat{\text{Var}}(X)}$$

2.5 Special Case: Regression Without an Intercept

Consider the constrained model where the intercept is restricted to zero:

$$Y_i = \beta_1 X_i + u_i$$

The fitted values are $\hat{Y}_i = \tilde{\beta}_1 X_i$, and the objective function to minimize is:

$$S(\tilde{\beta}_1) = \sum_{i=1}^n (Y_i - \tilde{\beta}_1 X_i)^2$$

Only One Normal Equation

Because there is no intercept parameter β_0 , we only differentiate with respect to $\tilde{\beta}_1$:

$$\frac{dS}{d\tilde{\beta}_1} = -2 \sum_{i=1}^n X_i (Y_i - \tilde{\beta}_1 X_i) = 0$$

Setting this to zero yields **only one Normal Equation**:

$$\sum_{i=1}^n X_i (Y_i - \tilde{\beta}_1 X_i) = 0 \quad \Longleftrightarrow \quad \tilde{\beta}_1 \sum_{i=1}^n X_i^2 = \sum_{i=1}^n X_i Y_i$$

The Resulting Slope Estimator

Solving the single normal equation for $\tilde{\beta}_1$ gives:

$$\tilde{\beta}_1 = \frac{\sum_{i=1}^n X_i Y_i}{\sum_{i=1}^n X_i^2}$$

2.6 Summary Comparison: With vs. Without Intercept

Property	With Intercept	Without Intercept
Model Specification	$Y_i = \beta_0 + \beta_1 X_i + u_i$	$Y_i = \beta_1 X_i + u_i$
Fitted Equation	$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$	$\hat{Y}_i = \tilde{\beta}_1 X_i$
Objective Function	$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$	$\min_{\tilde{\beta}_1} \sum_{i=1}^n (Y_i - \tilde{\beta}_1 X_i)^2$
Normal Equations	$\sum_{i=1}^n \hat{u}_i = 0$ and $\sum_{i=1}^n X_i \hat{u}_i = 0$	$\sum_{i=1}^n X_i \hat{u}_i = 0$
Intercept Estimator	$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$	None ($\tilde{\beta}_0 = 0$)
Slope Estimator	$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$	$\tilde{\beta}_1 = \frac{\sum_{i=1}^n X_i Y_i}{\sum_{i=1}^n X_i^2}$

3 Measures of Fit

Once we estimate an OLS regression line, we want to assess **how well the model fits the sample data**.

3.1 Sum of Squares Definitions (TSS, ESS, RSS)

We partition the total variation in the dependent variable Y_i into explained and unexplained components:

Total Sum of Squares (TSS)

Measures the **total sample variation** of Y_i around its sample mean $\bar{Y}$:

$$\text{TSS} = \sum_{i=1}^n (Y_i - \bar{Y})^2$$

Explained Sum of Squares (ESS)

Measures the variation in Y_i that is **explained by the regression model** (the variation of the fitted values $\hat{Y}_i$ around $\bar{Y}$):

$$\text{ESS} = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$$

Residual Sum of Squares (RSS / SSR)

Measures the **unexplained variation** in Y_i (the sum of squared prediction errors or residuals):

$$\text{RSS} = \sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

(Also frequently denoted as *SSR*, *Sum of Squared Residuals*).

3.2 Sum of Squares Decomposition

For an OLS regression model **with an intercept**, the total variation in Y decomposes into explained and residual variation:

$$\boxed{\text{TSS} = \text{ESS} + \text{RSS}}$$

Proof:

We can write each observed deviation from the sample mean as:

$$Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + (Y_i - \hat{Y}_i) = (\hat{Y}_i - \bar{Y}) + \hat{u}_i$$

Squaring both sides and summing over all $i = 1, \dots, n$:

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n \hat{u}_i^2 + 2 \sum_{i=1}^n (\hat{Y}_i - \bar{Y}) \hat{u}_i$$

Notice that $\hat{Y}_i - \bar{Y} = \hat{\beta}_1(X_i - \bar{X})$. Thus, the cross-product term is:

$$\sum_{i=1}^n (\hat{Y}_i - \bar{Y}) \hat{u}_i = \hat{\beta}_1 \sum_{i=1}^n (X_i - \bar{X}) \hat{u}_i = \hat{\beta}_1 \underbrace{\left(\sum_{i=1}^n X_i \hat{u}_i - \bar{X} \sum_{i=1}^n \hat{u}_i \right)}_{=0 \text{ by the OLS Normal Equations}} = 0$$

Since the cross-product is zero, we obtain:

$$\text{TSS} = \text{ESS} + \text{RSS}$$

Important Note: This clean decomposition $\text{TSS} = \text{ESS} + \text{RSS}$ requires $\sum_{i=1}^n \hat{u}_i = 0$, which holds **only when the regression includes an intercept**.

3.3 R^2

The R^2 is the **fraction of the sample variance of Y explained by X** :

$$\boxed{R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}}$$

3.4 SER (Standard Error of Regression)

The **Standard Error of the Regression (SER)**, denoted $s_{\hat{u}}$, estimates the standard deviation of the error term u_i :

$$\text{SER} = s_{\hat{u}} = \sqrt{\frac{\text{RSS}}{n-2}} = \sqrt{\frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2}$$

- **Degrees of Freedom ($n-2$):** We divide by $n-2$ because two degrees of freedom were used to estimate the two parameters β_0 and β_1 .
 - **Interpretation:** The SER measures the **typical size (spread) of the residuals** in the same units as the dependent variable Y .
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4 Sampling Properties of OLS

Now that we have derived the formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$, we study their statistical properties across random samples.

The Least Squares Assumptions

To analyze the sampling distribution of OLS estimators, we rely on three standard assumptions:

1. **Conditional Mean Zero (Exogeneity):** $\mathbb{E}[u_i | X_i] = 0$.
 2. **Independent and Identically Distributed (i.i.d.) Sampling:** $(X_1, Y_1), \dots, (X_n, Y_n)$ are i.i.d. draws from their joint distribution.
 3. **Finite 4th moments:** $0 < \mathbb{E}[X_i^4] < \infty$ and $0 < \mathbb{E}[Y_i^4] < \infty$.
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Key Algebraic Decomposition of $\hat{\beta}_1$

Before proving unbiasedness, consistency, and asymptotic normality, we express $\hat{\beta}_1$ in terms of the true parameter β_1 and the errors u_i .

Since $Y_i = \beta_0 + \beta_1 X_i + u_i$ and $\bar{Y} = \beta_0 + \beta_1 \bar{X} + \bar{u}$, subtracting gives:

$$Y_i - \bar{Y} = \beta_1(X_i - \bar{X}) + (u_i - \bar{u})$$

Substituting this into the OLS formula for $\hat{\beta}_1$:

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} \\ &= \frac{\sum_{i=1}^n (X_i - \bar{X}) [\beta_1 (X_i - \bar{X}) + (u_i - \bar{u})]}{\sum_{i=1}^n (X_i - \bar{X})^2} \\ &= \beta_1 \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sum_{i=1}^n (X_i - \bar{X})^2} + \frac{\sum_{i=1}^n (X_i - \bar{X})(u_i - \bar{u})}{\sum_{i=1}^n (X_i - \bar{X})^2}\end{aligned}$$

Because $\sum_{i=1}^n (X_i - \bar{X})\bar{u} = \bar{u} \sum_{i=1}^n (X_i - \bar{X}) = 0$, this simplifies to:

$$\boxed{\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n (X_i - \bar{X})u_i}{\sum_{i=1}^n (X_i - \bar{X})^2}}$$

4.1 Proof of OLS Unbiasedness

An estimator is **unbiased** if the mean of its sampling distribution equals the true parameter: $\mathbb{E}[\hat{\beta}_1] = \beta_1$ and $\mathbb{E}[\hat{\beta}_0] = \beta_0$.

1. Unbiasedness of the Slope ($\hat{\beta}_1$)

Condition on the entire sample of regressors $\mathbf{X} = (X_1, X_2, \dots, X_n)'$:

$$\mathbb{E}[\hat{\beta}_1 | \mathbf{X}] = \mathbb{E} \left[\beta_1 + \frac{\sum_{i=1}^n (X_i - \bar{X})u_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \middle| \mathbf{X} \right]$$

Since β_1 is a constant and the X_i terms are fixed when conditioning on $\mathbf{X}$:

$$\mathbb{E}[\hat{\beta}_1 | \mathbf{X}] = \beta_1 + \frac{\sum_{i=1}^n (X_i - \bar{X})\mathbb{E}[u_i | \mathbf{X}]}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Under the conditional mean zero assumption ($\mathbb{E}[u_i | \mathbf{X}] = 0$):

$$\mathbb{E}[\hat{\beta}_1 | \mathbf{X}] = \beta_1 + 0 = \beta_1$$

Applying the **Law of Iterated Expectations (LIE)** to obtain the unconditional expectation:

$$\mathbb{E}[\hat{\beta}_1] = \mathbb{E}[\mathbb{E}[\hat{\beta}_1 | \mathbf{X}]] = \mathbb{E}[\beta_1] = \beta_1$$

Thus, $\hat{\beta}_1$ is an **unbiased estimator** of β_1 .

2. Unbiasedness of the Intercept ($\hat{\beta}_0$)

Recall that $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$. Substituting $\bar{Y} = \beta_0 + \beta_1 \bar{X} + \bar{u}$:

$$\hat{\beta}_0 = (\beta_0 + \beta_1 \bar{X} + \bar{u}) - \hat{\beta}_1 \bar{X} = \beta_0 - (\hat{\beta}_1 - \beta_1) \bar{X} + \bar{u}$$

Taking conditional expectations:

$$\mathbb{E}[\hat{\beta}_0 | \mathbf{X}] = \beta_0 - \underbrace{(\mathbb{E}[\hat{\beta}_1 | \mathbf{X}] - \beta_1)}_{=0} \bar{X} + \underbrace{\mathbb{E}[\bar{u} | \mathbf{X}]}_{=0} = \beta_0$$

By the Law of Iterated Expectations:

$$\mathbb{E}[\hat{\beta}_0] = \mathbb{E}[\mathbb{E}[\hat{\beta}_0 | \mathbf{X}]] = \beta_0$$

Thus, $\hat{\beta}_0$ is also an **unbiased estimator** of β_0 .

4.2 Proof of OLS Consistency

An estimator is **consistent** if it converges in probability to the true population parameter as the sample size $n \rightarrow \infty$:

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 \quad \text{and} \quad \hat{\beta}_0 \xrightarrow{p} \beta_0$$

1. Consistency of the Slope ($\hat{\beta}_1$)

Divide both the numerator and the denominator of the estimation error by n :

$$\hat{\beta}_1 - \beta_1 = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}) u_i}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(u_i - \bar{u})}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

By the **Weak Law of Large Numbers (WLLN)** as $n \rightarrow \infty$: 1. **Numerator (Sample Covariance of X and u):**

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(u_i - \bar{u}) \xrightarrow{p} \text{Cov}(X_i, u_i)$$

Under Assumption 1 ($\mathbb{E}[u_i | X_i] = 0$), $\text{Cov}(X_i, u_i) = \mathbb{E}[(X_i - \mathbb{E}[X])u_i] = \mathbb{E}[(X_i - \mathbb{E}[X])\mathbb{E}[u_i | X_i]] = 0$. Therefore, the numerator converges in probability to 0.

2. Denominator (Sample Variance of X):

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \xrightarrow{p} \text{Var}(X_i) = \sigma_X^2$$

As long as there is population variation in X ($\sigma_X^2 > 0$), the denominator converges to a strictly positive constant.

By **Continuous Mapping Theorem / Slutsky's Theorem**:

$$\hat{\beta}_1 - \beta_1 \xrightarrow{p} \frac{0}{\sigma_X^2} = 0 \quad \Rightarrow \quad \boxed{\hat{\beta}_1 \xrightarrow{p} \beta_1}$$

2. Consistency of the Intercept ($\hat{\beta}_0$)

Since $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$, by the LLN $\bar{Y} \xrightarrow{p} \mathbb{E}[Y]$ and $\bar{X} \xrightarrow{p} \mathbb{E}[X]$. Because $\hat{\beta}_1 \xrightarrow{p} \beta_1$:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \xrightarrow{p} \mathbb{E}[Y] - \beta_1 \mathbb{E}[X]$$

From the true population model $Y_i = \beta_0 + \beta_1 X_i + u_i$, taking expectations gives $\mathbb{E}[Y] = \beta_0 + \beta_1 \mathbb{E}[X]$. Thus:

$$\hat{\beta}_0 \xrightarrow{p} (\beta_0 + \beta_1 \mathbb{E}[X]) - \beta_1 \mathbb{E}[X] = \beta_0 \quad \Rightarrow \quad \boxed{\hat{\beta}_0 \xrightarrow{p} \beta_0}$$

4.3 Proof of OLS Asymptotic Normality

An estimator is **asymptotically normal** if its distribution, properly centered and scaled by $\sqrt{n}$, converges in distribution to a normal distribution as $n \rightarrow \infty$:

$$\sqrt{n}(\hat{\beta}_1 - \beta_1) \xrightarrow{d} \mathcal{N}\left(0, \sigma_{\hat{\beta}_1}^2\right)$$

Proof for the Slope ($\hat{\beta}_1$):

Multiply the estimation error by $\sqrt{n}$:

$$\sqrt{n}(\hat{\beta}_1 - \beta_1) = \frac{\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \bar{X}) u_i}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

Let $v_i = (X_i - \mu_X) u_i$. Decomposing the numerator:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \bar{X}) u_i = \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mu_X) u_i - (\bar{X} - \mu_X) \frac{1}{\sqrt{n}} \sum_{i=1}^n u_i = \frac{1}{\sqrt{n}} \sum_{i=1}^n v_i + o_p(1)$$

Since $\bar{X} - \mu_X \xrightarrow{p} 0$ by the LLN and $\frac{1}{\sqrt{n}} \sum u_i = O_p(1)$ by the CLT, the second term vanishes in probability ($o_p(1)$).

1. **Numerator (Central Limit Theorem):** Because (X_i, Y_i) are i.i.d., the random variables $v_i = (X_i - \mu_X) u_i$ are i.i.d. with mean:

$$\mathbb{E}[v_i] = \mathbb{E}[(X_i - \mu_X) \mathbb{E}[u_i | X_i]] = 0$$

and variance $\sigma_v^2 = \text{Var}(v_i) = \mathbb{E}[(X_i - \mu_X)^2 u_i^2]$. By the **Central Limit Theorem (CLT)**:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n v_i \xrightarrow{d} \mathcal{N}(0, \sigma_v^2)$$

2. **Denominator (Law of Large Numbers):** By the **LLN**:

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \xrightarrow{p} \text{Var}(X_i) = \sigma_X^2$$

3. **Applying Slutsky's Theorem:** By Slutsky's Theorem:

$$\sqrt{n}(\hat{\beta}_1 - \beta_1) = \frac{\frac{1}{\sqrt{n}} \sum_{i=1}^n v_i}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2} \xrightarrow{d} \frac{\mathcal{N}(0, \sigma_v^2)}{\sigma_X^2} = \mathcal{N}\left(0, \frac{\sigma_v^2}{(\sigma_X^2)^2}\right)$$

where the asymptotic variance is:

$$\sigma_{\hat{\beta}_1}^2 = \frac{\text{Var}((X_i - \mu_X)u_i)}{(\text{Var}(X_i))^2} = \frac{\mathbb{E}[(X_i - \mu_X)^2 u_i^2]}{(\mathbb{E}[(X_i - \mu_X)^2])^2}$$

Key Takeaway: In large samples ($n \geq 30$), the sampling distribution of $\hat{\beta}_1$ is approximately normal:

$$\boxed{\hat{\beta}_1 \stackrel{a}{\sim} \mathcal{N}\left(\beta_1, \frac{\sigma_{\hat{\beta}_1}^2}{n}\right)}$$

This guarantees that standard t -statistics converge to $\mathcal{N}(0, 1)$, enabling valid hypothesis tests and confidence intervals.

5 Example: The Preston Curve

To connect the theoretical derivations of OLS, measures of fit (TSS, ESS, RSS, R^2 , SER), and asymptotic hypothesis testing to empirical practice, we examine one of the most famous empirical relationships in economics: the **Preston Curve** (Samuel Preston, 1975).

The Preston Curve describes the cross-country relationship between economic development (measured by GDP per capita) and population health (measured by life expectancy at birth).

5.1 Setting Up a Clean Do-File & Logging

In empirical research, we **never work from the interactive command window**. Every data transformation, cleaning step, and estimation must be written inside an auditable Stata do-file

(.do).

Following the workflow established in Recitation 1, every do-file should begin with standard initialization commands:

```
*=====
* Faiz Essa
* ECON UN3412: Econometrics
* Recitation 2: OLS in Stata (The Preston Curve)
* Companion .do file
*
* Required Subdirectories in Working Directory:
* - data/      (contains raw input data: wdi_gdp.csv, wdi_life_expectancy.csv)
* - output/    (stores intermediate .dta files and recitation2.log)
* - images/    (stores exported plots: preston_linear.pdf, preston_log.pdf)
*=====

* 0. Environment Setup & Logging
capture log close           // Close any open logs
clear all                  // Clear memory
set more off               // Disable pause prompts during long output

* Set working directory to project root (adjust path as needed)
cd "/Users/faizessa/Library/CloudStorage/Dropbox/Mac/Documents/ta/ECONUN3412"

* Open log file to record commands and console output
log using "output/recitation2.log", replace text
```

Why Logging Matters: The `log using ..., replace text` command creates a plain-text transcript of everything displayed in Stata's Results window. This ensures your empirical analysis is 100% auditable and reproducible. When the script finishes, we close the session with `log close`.

5.2 Importing the Raw Datasets

Economists frequently need to combine data from multiple sources. In our `data/` directory, we have two separate World Bank CSV files for the year 2019: 1. `data/wdi_gdp.csv`: Contains country codes, country names, geographic regions, income groups, and GDP per capita (`gdppc`). 2. `data/wdi_life_expectancy.csv`: Contains country codes, country names, life expectancy at birth in years (`life_exp`), and infant mortality (`infant_mort`).

We import each CSV using `import delimited` and save them as native Stata `.dta` files:

```

* 1. Import GDP per capita CSV and save as Stata .dta
import delimited "data/wdi_gdp.csv", clear
describe
save "output/wdi_gdp.dta", replace

* 2. Import Life Expectancy CSV and save as Stata .dta
import delimited "data/wdi_life_expectancy.csv", clear
describe
save "output/wdi_life_expectancy.dta", replace

```

5.3 Merging Datasets in Stata (merge 1:1)

To merge two datasets, we need a **primary key**—a variable that uniquely identifies observations in both files. Here, both datasets share `country_code` (the 3-letter ISO code, e.g., USA, FRA, KEN).

In Stata, the dataset currently open in memory is called the **master dataset**, while the file stored on disk is called the **using dataset**:

```

* Load master dataset
use "output/wdi_gdp.dta", clear

* Perform a one-to-one match merge
merge 1:1 country_code using "output/wdi_life_expectancy.dta"

```

Understanding the System Variable `_merge`

Stata automatically creates a diagnostic variable named `_merge` with three possible values:

Value	Label	Meaning
<code>_merge == 1</code>	master only	Observation appeared in master (<code>wdi_gdp</code>), but not in using
<code>_merge == 2</code>	using only	Observation appeared in using (<code>wdi_life_expectancy</code>), but not in master
<code>_merge == 3</code>	matched	Observation matched successfully in both datasets

Stata prints the following diagnostic table to the console:

Result	# of obs.	
not matched	6	
from master	0	(<code>_merge==1</code>)
from using	6	(<code>_merge==2</code>)
matched	211	(<code>_merge==3</code>)

Auditing and Cleaning Post-Merge

Notice that 6 countries/territories were in the life expectancy file but lacked GDP data for 2019 (`_merge == 2`). For our regression analysis, we keep only observations with complete data across both sources (`_merge == 3`), drop the temporary `_merge` indicator, and save the consolidated dataset:

```
* Inspect the breakdown
tabulate _merge

* Keep only matched observations across both files
keep if _merge == 3

* Drop the temporary diagnostic variable
drop _merge

* Save the clean merged dataset
save "output/wdi_preston_merged.dta", replace
```

5.4 Evaluating OLS Assumptions: The “Linearity” Assumption

Recall our baseline linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

Under **Least Squares Assumption 1** (Conditional Mean Zero / Exogeneity):

$$\mathbb{E}[u_i | X_i] = 0 \quad \implies \quad \mathbb{E}[Y_i | X_i] = \beta_0 + \beta_1 X_i$$

This equation states that the **Population Regression Function (PRF)** is strictly **linear in the regressor X** :

$$\frac{d\mathbb{E}[Y | X]}{dX} = \beta_1 = \text{constant}$$

The Economic Problem with Linearity in Health & Income

In economics, assuming a constant marginal effect β_1 across all income levels is deeply unrealistic:

* **In Low-Income Countries:** Moving from \$500 to \$1,500 of GDP per capita allows a nation to establish basic public health infrastructure: clean drinking water, sewage treatment, childhood vaccinations, and rudimentary medical clinics. This yields a **massive increase** in life expectancy (often 10 to 15 years). * **In High-Income Countries:** Moving from \$50,000 to \$51,000 of GDP per capita provides marginal comforts, but basic diseases are already eradicated. Extending life further requires expensive treatments for chronic late-life illnesses, yielding only **marginal gains** in longevity.

This phenomenon is known as **diminishing marginal returns to income**. If the true relationship is concave rather than linear, fitting a straight OLS line will violate $\mathbb{E}[u | X] = 0$: the residuals $\hat{u}_i$ will be systematically positive in some regions and systematically negative in others.

5.5 Plotting the Data: Testing the Linearity Assumption

Because GDP per capita is measured in raw dollars (running from \$234 to \$193,747), a \$1 increase is negligible. We rescale GDP per capita into **thousands of USD** so that our slope coefficients will have practical, interpretable magnitudes:

```
* Rescale GDP per capita into thousands of USD
generate gdppc_k = gdppc / 1000
label variable gdppc_k "GDP per capita (thousands of USD)"
label variable life_exp "Life expectancy at birth (years)"

* Summary statistics
summarize life_exp gdppc_k
```

Stata's summary table displays:

Variable	Obs	Mean	Std. Dev.	Min	Max
-----+-----					
life_exp	211	73.09047	7.963422	31.53	86.15
gdppc_k	211	19.10102	27.46402	.23431	193.7467

Now we construct a two-way graph overlaying the raw scatterplot with Stata's linear OLS line of best fit (`lfit`):

```

twoway (scatter life_exp gdppc_k, mcolor(navy%70) msize(small)) ///
      (lfit life_exp gdppc_k, lcolor(cranberry) lwidth(medium)), ///
      title("Preston Curve: Testing the Linearity Assumption", size(medium)) ///
      subtitle("Life Expectancy vs. GDP per Capita (2019)", size(small)) ///
      xtitle("GDP per Capita (Thousands of Current USD)") ///
      ytitle("Life Expectancy at Birth (Years)") ///
      legend(order(1 "Countries" 2 "Linear OLS Fit")) ///
      scheme(s2color) graphregion(color(white))

graph export "images/preston_linear.pdf", replace

```

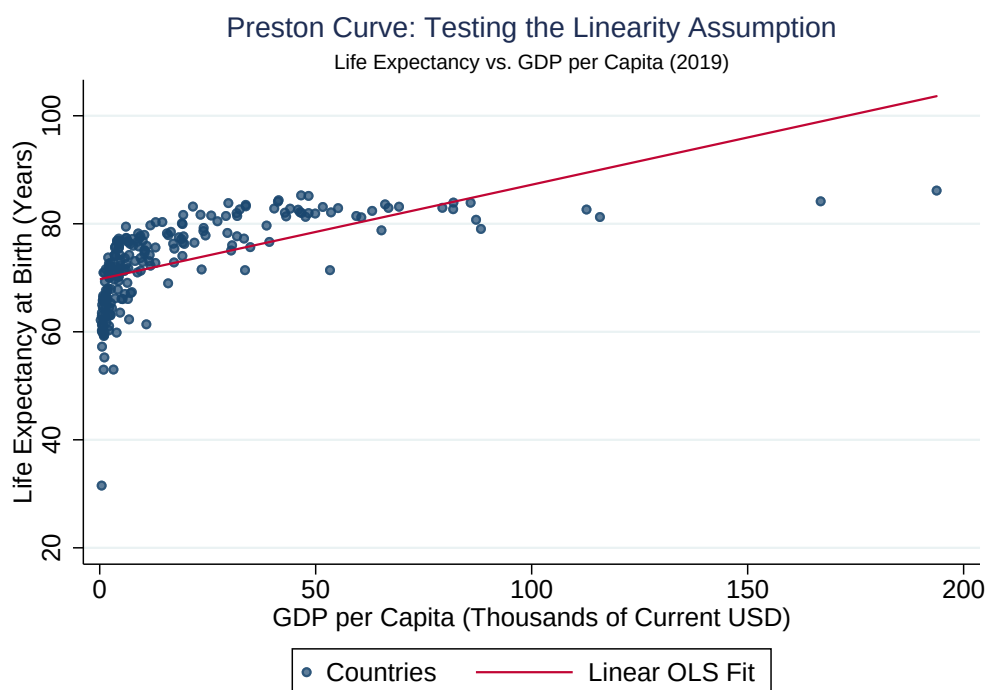


Figure 2: Preston Curve with Linear OLS Fit. Notice the pronounced concavity: the linear specification systematically underpredicts life expectancy for middle-income countries and fails to capture the steep slope at low incomes.

Visual Diagnostic

Looking at Figure 2, the violation of linearity is glaring: 1. The true data curve rises steeply between \$0k and \$15k, then bends sharply and flattens out beyond \$30k. 2. The straight OLS line **severely underpredicts** life expectancy for emerging and middle-income nations (e.g., countries with \$10k–\$30k income clustered far *above* the line). 3. The linear line **overpredicts** life expectancy for the poorest nations and the very wealthiest nations.

The linear model is clearly **misspecified**.

5.6 Choosing an Alternative: The Linear-Log (Semi-Elasticity) Specification

To model diminishing marginal returns, economists transform the regressor using the **natural logarithm**:

$$\text{LifeExp}_i = \alpha_0 + \alpha_1 \ln(\text{GDPpc}_i) + v_i$$

(Note: In econometrics, when Y is in levels and X is in logarithms, this specification is formally called a **linear-log model**.)

Step 1: Why Natural Logarithms Model Diminishing Returns

Recall from calculus that the derivative of $\ln(X)$ is:

$$\frac{d}{dX} \ln(X) = \frac{1}{X}$$

* When X is small (a low-income country), $\frac{1}{X}$ is very large: an additional dollar causes a steep rise in $\ln(X)$. * As X grows large (a high-income country), $\frac{1}{X}$ approaches zero: an additional dollar barely changes $\ln(X)$.

By taking the logarithm of income, we compress large dollar values and stretch out small dollar values, naturally capturing the concavity observed in the Preston Curve.

Step 2: Deriving the Semi-Elasticity (Step-by-Step)

1. **The Fundamental Log Property:** For any small relative change $\frac{\Delta X}{X}$, the change in the natural logarithm approximates the **fractional (proportional) change**:

$$\Delta \ln(X) = \ln(X + \Delta X) - \ln(X) \approx \frac{\Delta X}{X}$$

2. **Connecting to the Regression Slope:** Taking differences across our regression equation $\widehat{Y} = \hat{\alpha}_0 + \hat{\alpha}_1 \ln(X)$:

$$\Delta \widehat{Y} = \hat{\alpha}_1 \cdot \Delta \ln(X) \approx \hat{\alpha}_1 \cdot \left(\frac{\Delta X}{X} \right)$$

3. **Converting Fractional Change to Percentage Change:** Recall that a **percentage change** is simply the fractional change multiplied by 100:

$$\% \Delta X = 100 \times \left(\frac{\Delta X}{X} \right) \implies \frac{\Delta X}{X} = \frac{\% \Delta X}{100}$$

Substituting this into our equation for $\widehat{\Delta Y}$:

$$\widehat{\Delta Y} \approx \hat{\alpha}_1 \left(\frac{\% \Delta X}{100} \right) = \left(\frac{\hat{\alpha}_1}{100} \right) \times (\% \Delta X)$$

Step 3: The Practical Rule of Thumb

From this derivation, we obtain the universal interpretation rule for linear-log models:

The Linear-Log Interpretation Rule: A 1% increase in X ($\% \Delta X = 1$) is associated with an expected change in Y of:

$$\widehat{\Delta Y} \approx \frac{\hat{\alpha}_1}{100} \text{ units of } Y$$

More generally, a $p\%$ increase in X is associated with an expected change of $\left(\frac{\hat{\alpha}_1}{100} \right) \times p$ units of Y .

Step 4: Economic Intuition

Notice what this specification implies economically: **an equal percentage increase in income yields an equal absolute gain in life expectancy across all countries.**

Consider a 10% economic growth rate: * **In a low-income country (GDP = \$1,000):** A 10% increase is only +\$100. * **In a wealthy country (GDP = \$50,000):** A 10% increase is +\$5,000.

Under the linear-log model, both countries achieve the **exact same predicted gain in life expectancy:**

$$\widehat{\Delta \text{LifeExp}} \approx \left(\frac{\hat{\alpha}_1}{100} \right) \times 10 = 0.10 \times \hat{\alpha}_1 \text{ years}$$

This directly mirrors economic reality: it takes 50 **times more dollars** in a wealthy nation (\$5,000 vs. \$100) to achieve the same additional year of life expectancy because basic health needs are already satisfied.

Let us generate the log variable in Stata and plot the transformed data:

```
* Create natural logarithm of raw GDP per capita
generate log_gdppc = log(gdppc)
label variable log_gdppc "Log GDP per capita"

* Plot scatterplot with Log-Linear OLS fit
twoway (scatter life_exp log_gdppc, mcolor(navy%70) msize(small)) ///
      (lfit life_exp log_gdppc, lcolor(cranberry) lwidth(medium)), ///
```

```

title("Preston Curve: Log-Linear Specification", size(medium)) ///
subtitle("Life Expectancy vs. Log GDP per Capita (2019)", size(small)) ///
xtitle("Natural Log of GDP per Capita, ln(GDP)") ///
yttitle("Life Expectancy at Birth (Years)") ///
legend(order(1 "Countries" 2 "Log-Linear OLS Fit")) ///
scheme(s2color) graphregion(color(white))

graph export "images/preston_log.pdf", replace

```

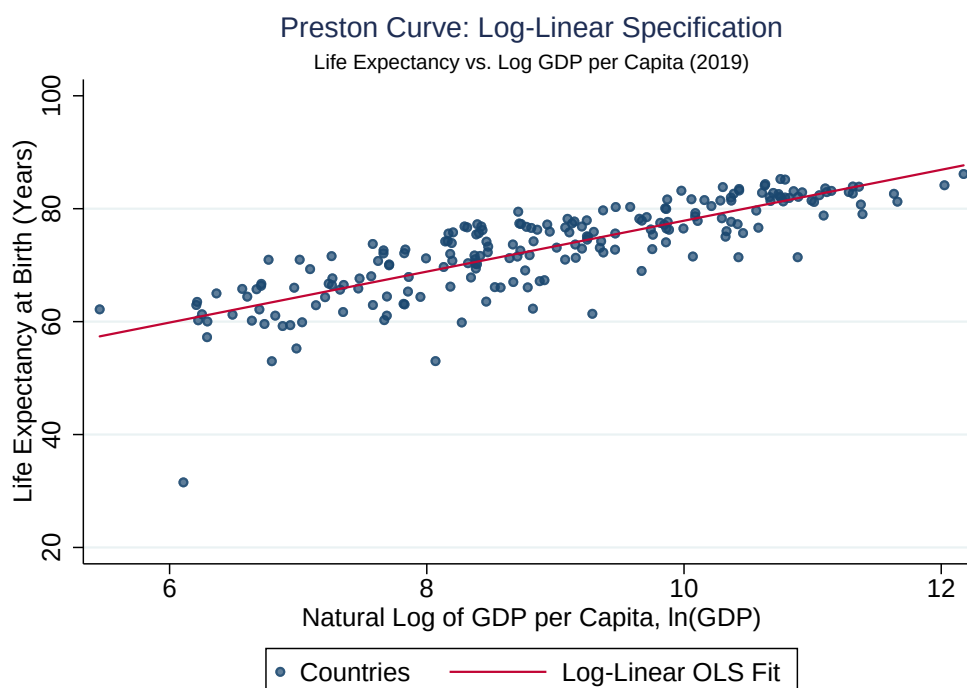


Figure 3: Preston Curve with Log-Linear OLS Fit. Transforming GDP per capita into natural logarithms straightens the relationship into an extraordinarily linear cloud of points.

As seen in Figure 3, taking the natural logarithm linearizes the relationship: the country observations now scatter symmetrically around the straight line across the entire distribution of income.

5.7 Running the OLS Regressions in Stata

We now estimate both specifications using Stata's **regress** command:

```

* Model 1: Baseline Linear Model
regress life_exp gdppc_k

```

```
* Model 2: Log-Linear (Semi-Elasticity) Model
regress life_exp log_gdppc
```

Stata Regression Output: Model 1 (Linear)

```
. regress life_exp gdppc_k
```

Source		SS		df	MS	Number of obs	=	211
-----+-----						F(1, 209)	=	119.43
Model		4842.75969		1	4842.75969	Prob > F	=	0.0000
Residual		8474.61845		209	40.5484137	R-squared	=	0.3636
-----+-----						Adj R-squared	=	0.3606
Total		13317.3781		210	63.4160864	Root MSE	=	6.3678
-----+-----								
life_exp		Coef.		Std. Err.	t	P> t	[95% Conf. Interval]	
-----+-----								
gdppc_k		.1748528		.0159997	10.93	0.000	.1433113	.2063944
_cons		69.75061		.5343885	130.52	0.000	68.69712	70.80409
-----+-----								

Stata Regression Output: Model 2 (Log-Linear)

```
. regress life_exp log_gdppc
```

Source		SS		df		MS		Number of obs	=	211
-----+-----								F(1, 209)	=	456.83
Model		9137.11907		1		9137.11907		Prob > F	=	0.0000
Residual		4180.25907		209		20.0012396		R-squared	=	0.6861
-----+-----								Adj R-squared	=	0.6846
Total		13317.3781		210		63.4160864		Root MSE	=	4.4723

life_exp		Coef.		Std. Err.		t		P> t		[95% Conf. Interval]
-----+-----										
log_gdppc		4.509862		.2110022		21.37		0.000		4.093897 4.925828
_cons		32.78044		1.910945		17.15		0.000		29.01324 36.54764

5.8 Deconstructing and Interpreting the Stata Regression Table

Every cell in Stata's regression table maps directly to the theoretical formulas derived earlier in this handout.

1. The ANOVA Sum of Squares Table (Top-Left)

- **Model SS (ESS):** The variation in life expectancy explained by the regression line:

$$\text{ESS}_{\text{linear}} = 4,842.76, \quad \text{ESS}_{\text{log}} = 9,137.12$$

- **Residual SS (RSS / SSR):** The unexplained variation (sum of squared residuals):

$$\text{RSS}_{\text{linear}} = 8,474.62, \quad \text{RSS}_{\text{log}} = 4,180.26$$

- **Total SS (TSS):** The total sample variation of Y around $\bar{Y}$:

$$\text{TSS} = \sum_{i=1}^n (Y_i - \bar{Y})^2 = 13,317.38$$

Notice that TSS is **identical** in both models (13,317.38) because the dependent variable Y_i is identical.

- **Sum of Squares Identity:** As proved in Section 3.2, for any regression with an intercept:

$$\text{TSS} = \text{ESS} + \text{RSS}$$

In Model 1: $4,842.76 + 8,474.62 = 13,317.38$.

- **Degrees of Freedom (df):**
 - Model: $k = 1$ slope parameter.
 - Residual: $n - 2 = 211 - 2 = 209$.
 - Total: $n - 1 = 211 - 1 = 210$.

2. Summary Fit Statistics (Top-Right)

- **Sample Size (Number of obs):** $n = 211$ countries.
- **R^2 (R-squared):** The fraction of total variation in life expectancy explained by income:

$$R^2_{\text{linear}} = \frac{\text{ESS}}{\text{TSS}} = \frac{4,842.76}{13,317.38} = \mathbf{0.3636}$$

$$R^2_{\text{log}} = \frac{\text{ESS}}{\text{TSS}} = \frac{9,137.12}{13,317.38} = \mathbf{0.6861}$$

> **Key Takeaway:** Simply transforming GDP per capita into logs nearly **doubles the explanatory power** of the regression (R^2 jumps from 36.4% to 68.6%).

- **Standard Error of the Regression (Root MSE = SER = $s_{\hat{u}}$):** Measures the typical size of the residual prediction error in the original units of Y (years):

$$\text{SER}_{\text{linear}} = \sqrt{\frac{\text{RSS}}{n-2}} = \sqrt{\frac{8,474.62}{209}} = \mathbf{6.368} \text{ years}$$

$$\text{SER}_{\text{log}} = \sqrt{\frac{\text{RSS}}{n-2}} = \sqrt{\frac{4,180.26}{209}} = \mathbf{4.472} \text{ years}$$

The log specification reduces the standard prediction error by nearly **2 full years** of life.

3. Parameter Estimates & Inference (Bottom Table)

Model 1: Baseline Linear Model

The estimated OLS sample regression line is:

$$\widehat{\text{LifeExp}}_i = 69.75 + 0.1749 \times \text{GDPpc_k}_i$$

- **Slope Coefficient ($\hat{\beta}_1 = 0.1749$):**
 - **Economic Interpretation:** An increase of \$1,000 in GDP per capita is associated with an expected increase of 0.175 **years** (approximately 2.1 months) in life expectancy at birth.
 - **Standard Error ($\text{SE}(\hat{\beta}_1) = 0.0160$):** Measures the sampling variability of the estimator across repeated random samples.
 - **t -Statistic & p -Value:**

$$t = \frac{\hat{\beta}_1 - 0}{\text{SE}(\hat{\beta}_1)} = \frac{0.17485}{0.01600} = 10.93$$

Since $|t| = 10.93 \gg 1.960$ and $p < 0.001$, we **reject** $H_0 : \beta_1 = 0$ at any standard significance level ($\alpha = 0.01, 0.05, 0.10$).

- **95% Confidence Interval:** $[0.143, 0.206]$. We are 95% confident that the true population increase in life expectancy per \$1,000 lies between 0.143 and 0.206 years.
- **Intercept Coefficient ($\hat{\beta}_0 = 69.75$):**
 - **Interpretation:** Predicts an average life expectancy of 69.75 years for a hypothetical country with GDP = \$0.
 - **Econometric Caution:** No country in the sample has zero income (Min = \$234). The intercept represents a pure out-of-sample extrapolation to anchor the line of best fit.

Model 2: Log-Linear Model

The estimated OLS sample regression line is:

$$\widehat{\text{LifeExp}}_i = 32.78 + 4.510 \times \ln(\text{GDPpc}_i)$$

- **Slope Coefficient** ($\hat{\alpha}_1 = 4.510$):

- **Economic Interpretation (1% Increase):** A 1% increase in GDP per capita is associated with an expected increase in life expectancy of:

$$\frac{\hat{\alpha}_1}{100} = \frac{4.510}{100} = \mathbf{0.0451} \text{ years } (\approx 16.5 \text{ days})$$

- **Interpretation (10% Increase):** A 10% increase in GDP per capita is associated with ≈ 0.45 additional years (≈ 5.4 months) of life.
- **Interpretation (Doubling of Income):** Doubling a country's GDP per capita represents a $\approx 70\%$ increase ($\ln(2) \approx 0.693$). Therefore, doubling income is associated with:

$$\ln(2) \times 4.510 \approx 0.693 \times 4.510 = \mathbf{3.13} \text{ additional years}$$

- **Statistical Significance:** $t = 21.37$ ($p < 0.0001$), with a tight 95% confidence interval of $[4.094, 4.926]$.

5.9 Hypothesis Testing with Stata's `test` Command

In addition to testing the default null hypothesis $H_0 : \beta_1 = 0$ reported in the output table, economists often test **specific non-zero economic hypotheses**.

For example, suppose a health policy researcher hypothesizes that an additional \$1,000 of income increases life expectancy by **half a year** ($H_0 : \beta_1 = 0.5$ vs. $H_1 : \beta_1 \neq 0.5$).

Immediately after running `regress`, we test this hypothesis using Stata's post-estimation `test` command:

```
regress life_exp gdppc_k
test gdppc_k = 0.5
```

Stata returns:

```
( 1)  gdppc_k = .5

      F( 1, 209) = 412.98
      Prob > F = 0.0000
```

Understanding the Stata test Output

1. **Relationship between F and t :** For a single linear restriction on one parameter, the F -statistic is mathematically identical to the squared t -statistic:

$$t = \frac{\hat{\beta}_1 - 0.5}{\text{SE}(\hat{\beta}_1)} = \frac{0.17485 - 0.5}{0.0159997} = \frac{-0.32515}{0.01600} = -20.322$$

$$F = t^2 = (-20.322)^2 = \mathbf{412.98}$$

2. **Decision:** Because $F = 412.98$ yields a p -value < 0.0001 , we **decisively reject** $H_0 : \beta_1 = 0.5$ at the 1% significance level. The actual return to \$1,000 of income across the whole sample is substantially lower than half a year.